

Anne-Marie George

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I work on preference learning, computational social choice, and fairness in AI — developing methods for eliciting, aggregating, and reasoning about preferences, with an eye on how these choices shape fairness in collective decision-making. My work is primarily theoretical, drawing on combinatorial optimisation, graph theory, logic, and learning theory.

I am a Principal Investigator at Integreat, Norway's Centre for Trustworthy AI, on the Fairness & Explainability theme, and at TRUST — Centre for Trustworthy AI, in the research area *AI for Flourishing*.

Research

Preferences & collective decision-making

Methods for learning, eliciting, and aggregating preferences, with applications spanning recommender systems and language model alignment.

Fairness in AI

How fairness should be conceptualised in preference-based AI systems, and what that means for collective decision-making in practice.

Methods

Primarily theoretical work drawing on combinatorial optimisation, graph theory, logic, learning theory, and online learning — as well as computational social choice, approval voting, liquid democracy, stable matching, and preference modelling more broadly.

Teaching & Service

Teaching

Lecturer — Logiske metoder
IN1150, University of Oslo, Spring 2025

Lecturer — Reinforcement Learning and Decision Making Under Uncertainty
IN-STK5100 / IN-STK9100, University of Oslo, Spring 2022 and 2023

Lecturer — Adaptive Methods for Data-Based Decisions
IN-STK5000 / IN-STK9000, University of Oslo, Autumn 2022

Lecturer — Mallows Model Tutorial
Preference Statistics Summer School, July 2025

Lecturer — Reinforcement Learning Tutorial
23rd Geilo Winter School on Computational Statistics, January 2023

PhD Co-Supervision

Lorenzo Zuccato (since 2024)

Meirav Segal (PhD completed 2025)

Thomas Kleine Buening (PhD completed 2024)

Master's Supervision

“Actionable Recourse Over Time”, 2025/26

“Preference Models for Decision Making: The Moral Machine Case”, 2023/24

“College Admissions with Quotas and Bonus Points”, 2022/23

Talks & Outreach

“Fairness by Design — Considering Fairness in Algorithmic Decision Making”, Data Science Day, dScience Centre, University of Oslo, 2024

Service

Local organising committee, International Workshop on Statistical Modelling (IWSM), Oslo 2026

Reviewer / PC member: AAAI, AISTATS, IJCAI, SCALA

Member of the Green Council, Department of Informatics, University of Oslo (since 2023)

Awards

Outstanding Paper Award, ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO) 2022

Best Paper Award, Cooperative AI Workshop @ NeurIPS 2021

Best Poster Award, COST IC1205 Summer School on Computational Social Choice, 2016

Publications

2026

Prognostic biomarker discovery in pancreatic cancer through hybrid ensemble feature selection and multi-omics data JOURNAL
John Zobolas, **Anne-Marie George**, Alberto López, Sebastian Fischer 0004, Marc Becker, Tero Aittokallio
BioData Min.

Cycles in Liquid Democracy: A Game-Theoretic Justification CONFERENCE
Markus Brill, Rachael Colley, **Anne-Marie George**, Grzegorz Lisowski, Georgios Papasotiropoulos, Ulrike Schmidt-Kraepelin
IJCAI/ECAI

Probably Correct Optimal Stable Matching under Two-Sided Uncertainty PREPRINT
Andreas Athanasopoulos, **Anne-Marie George**, Christos Dimitrakakis
CoRR

2025

Probably Correct Optimal Stable Matching for Two-Sided Market Under Uncertainty CONFERENCE
Andreas Athanasopoulos, **Anne-Marie George**, Christos Dimitrakakis
AAMAS

On Middle Grounds for Preference Statements CONFERENCE
Anne-Marie George, Ana Ozaki
IJCAI

Probably Correct Optimal Stable Matching for Two-Sided Markets Under Uncertainty PREPRINT
Andreas Nikolaos Athanasopoulos, **Anne-Marie George**, Christos Dimitrakakis
CoRR

Optimizing Prognostic Biomarker Discovery in Pancreatic Cancer Through Hybrid Ensemble Feature Selection and Multi-Omics Data PREPRINT
John Zobolas, **Anne-Marie George**, Alberto López, Sebastian Fischer 0004, Marc Becker, Tero Aittokallio
CoRR

2024

Minimal Macro-Based Rewritings of Formal Languages: Theory and Applications in Ontology Engineering (and Beyond) CONFERENCE
Christian Kindermann, **Anne-Marie George**, Bijan Parsia, Uli Sattler
AAAI

Eliciting Kemeny Rankings CONFERENCE
Anne-Marie George, Christos Dimitrakakis
AAAI

Threshold Recourse for Dynamic Allocations CONFERENCE
Meirav Segal, **Anne-Marie George**, Christos Dimitrakakis
EWAF

Better Luck Next Time: About Robust Recourse in Binary Allocation Problems CONFERENCE
Meirav Segal, **Anne-Marie George**, Ingrid Chieh Yu, Christos Dimitrakakis
xAI (3)

2023

Policy Fairness and Unknown Bias Dynamics in Sequential Allocations CONFERENCE
Meirav Segal, **Anne-Marie George**, Christos Dimitrakakis
EAAMO

2022

The excess method: a multiwinner approval voting procedure to allocate wasted votes JOURNAL
Steven J. Brams, Markus Brill, **Anne-Marie George**
Soc. Choice Welf.

Liquid Democracy with Ranked Delegations CONFERENCE
Markus Brill, Théo Delemazure, **Anne-Marie George**, Martin Lackner, Ulrike Schmidt-Kraepelin
AAAI

On Meritocracy in Optimal Set Selection CONFERENCE
Thomas Kleine Buening, Meirav Segal, Debabrota Basu, **Anne-Marie George**, Christos Dimitrakakis
EAAMO

Interactive Inverse Reinforcement Learning for Cooperative Games CONFERENCE
Thomas Kleine Büning, **Anne-Marie George**, Christos Dimitrakakis
ICML

Single-Peaked Opinion Updates CONFERENCE
Robert Bredereck, **Anne-Marie George**, Jonas Israel, Leon Kellerhals
IJCAI

Simulating University Application Data for Fair Matchings CONFERENCE
Meirav Segal, **Anne-Marie George**, Christos Dimitrakakis
NAIS

2020

Assigning and Scheduling Service Visits in a Mixed Urban/Rural Setting JOURNAL

Mark Antunes, Vincent Armant, Kenneth N. Brown, Daniel A. Desmond, Guillaume Escamocher, **Anne-Marie George**, Diarmuid Grimes, Mike O'Keeffe, Yiqing Lin, Barry O'Sullivan, Cemalettin Ozturk, Luis Quesada 0001, Mohamed Siala 0002, Helmut Simonis, Nic Wilson

Int. J. Artif. Intell. Tools

2019

Preference inference based on lexicographic and Pareto models THESIS

Anne-Marie George

University College Cork, Republic of Ireland

2017

Efficient Inference and Computation of Optimal Alternatives for Preference Languages Based On Lexicographic Models CONFERENCE

Nic Wilson, **Anne-Marie George**

IJCAI

Preference Inference Based on Hierarchical and Simple Lexiographic Models JOURNAL

Nic Wilson, **Anne-Marie George**, Barry O'Sullivan

FLAP

2016

Towards Fast Algorithms for the Preference Consistency Problem Based on Hierarchical Models CONFERENCE

Anne-Marie George, Nic Wilson, Barry O'Sullivan

IJCAI

Preference Inference Based on Pareto Models CONFERENCE

Anne-Marie George, Nic Wilson

SUM

2015

Computation and Complexity of Preference Inference Based on Hierarchical Models CONFERENCE

Nic Wilson, **Anne-Marie George**, Barry O'Sullivan

IJCAI

The Comparison of Multi-objective Preference Inference Based on Lexicographic and Weighted Average Models CONFERENCE

Anne-Marie George, Abdul Razak, Nic Wilson

ICTAI